4 A d.c. power supply of e.m.f. 8.7 V and negligible internal resistance is connected by two identical connecting wires to three identical filament lamps, as shown in Fig. 4.1.

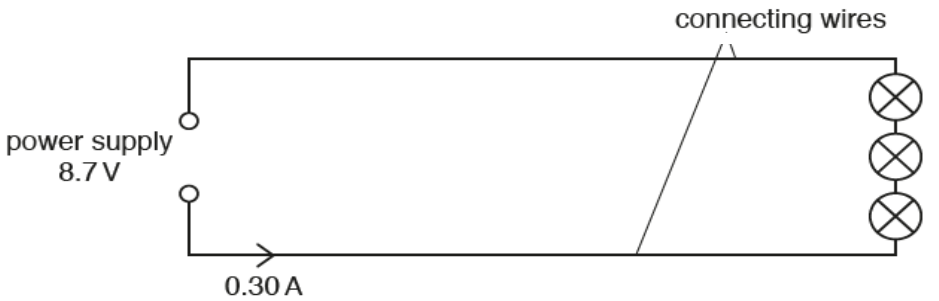

Fig. 4.1

The power supply provides a current of 0.30 A to the circuit.

The *I* – *V* characteristic for one of the lamps is shown in Fig. 4.2.

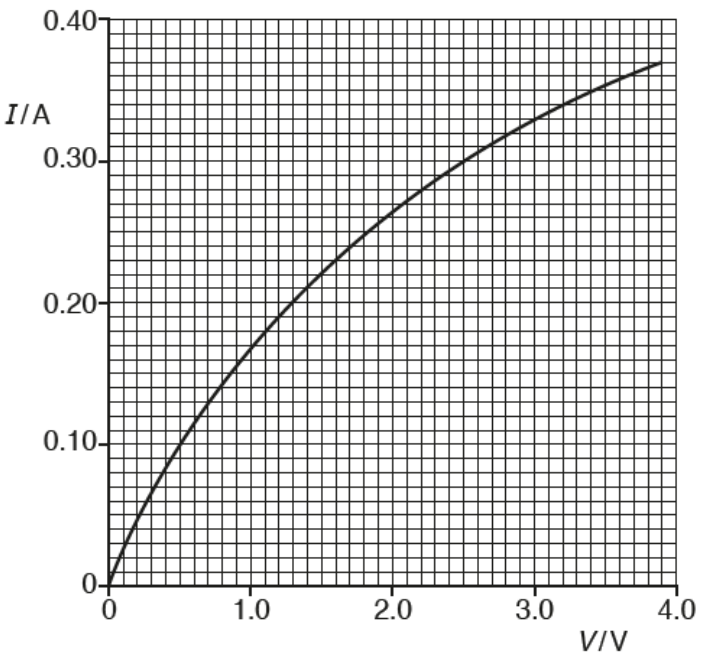

Fig. 4.2

(a) Show that the resistance of each connecting wire in Fig. 4.1 is 2.0 Ω.

[3]

(b) The resistance of the connecting wires does not vary with temperature.
On Fig. 4.2, sketch the *I*–*V* characteristic for **one** of the connecting wires. [2]

(c) Calculate the power loss in one of the connecting wires.

power loss = ... W [1]

(d) Some data for the connecting wires are given below.

cross-sectional area = 0.40 mm^2
resistivity = 1.7×10^{-8} Ω m
number density of free electrons = 8.5×10^{28} m^{-3}

Calculate

(i) the length of one of the connecting wires,

length = ... m [2]

(ii) the drift speed of a free electron in the connecting wires.

drift speed = ... m s^{-1} [2]

5 **(a)** A particle has mass m, charge $+q$ and speed v.